

MatRs - CP2:-
(PYQs):-

CG - 2025:-

CO5:-

	c_1	c_2	c_3	c_4
R_1	3	5	4	2
R_2	5	6	2	4
R_3	4	1	3	0
R_4	3	3	5	4

- Saddle:-

Row Minimum $\rightarrow$ Row 1: 2 Row 3: 0
 Row 2: 2 Row 4: 3
 Max $\rightarrow$ min = 3

Col Maximum $\rightarrow$ Col 1: 5 Col 3: 5
 Col 2: 6 Col 4: 4

Mini of them = 4

- Principle of Dominance:-

Row: Avg of Row R_1 & $R_2 = (4, 5.5, 3, 3)$.

$R_3 (4, 1, 3, 0)$.

Elements of $R_3 \leq$ Avg of R_1 & R_2 .

$\downarrow$

Eliminate R_3 .

Col:-

$c_2 \geq c_1 \rightarrow$ for all.

Player A $\rightarrow$ Minimize payoff.
 Eliminate c_2 .

	c_1	c_3	c_4
R_1	3	4	2
R_3	4	3	0
R_4	3	5	4

$R_4' \geq R_1' \rightarrow$ Eliminate R_1' $\rightarrow$ Eliminate c_4'
 $c_4'' \geq \left(\frac{c_1'' + c_3''}{2}\right)$

R_2, R_4 ; C_1, C_3

$$\begin{pmatrix} 5 & 2 \\ 3 & 5 \end{pmatrix}$$

$$5p + 3(1-p) = 2p + 5(1-p)$$

$$2p + 3 = 2p - 5p + 5$$

$$5p = 2$$

$$p = \frac{2}{5}$$

$$V = 5 \cdot \left(\frac{2}{5}\right) + 3 \cdot \left(\frac{3}{5}\right) = \frac{19}{5} = 3.8$$

$$\left(0, \frac{2}{5}, 0, \frac{3}{5}\right)$$

$$\left(\frac{3}{5}, 0, \frac{2}{5}, 0\right)$$

$V = 3.8$

(b)

Player A	Player B	
	B_1	B_2
A_1	1	2
A_2	4	5
A_3	9	-7
A_4	-3	-4
A_5	2	1

$A_2 > A_1, A_4, A_5$

(Player A.)
(Maximize)

Eliminate A_1, A_4, A_5 .

	B_1	B_2
A_2	4	5
A_3	9	-7

Player B $\rightarrow B_1$: Probability q ,
 $\rightarrow B_2$: Probability $(1-q)$

$$A_2: 4q + 5(1-q) = -q + 5,$$

$$A_3: 9q + (-7)(1-q) = 16q - 7,$$

from $q=0$ to $q=1$.

$$A_2 = 5$$

$$A_2 = 4$$

$$A_3 = -7$$

$$A_3 = 9$$

$$-q + 5 = 16q - 7.$$

$$17q = 12.$$

$$q = \frac{12}{17}$$

$$(1-q) \rightarrow \frac{5}{17}.$$

$$u = -\frac{12}{17} + 5 = \frac{43}{17}.$$

write Player A:

$$4p + 9(1-p) = 5p - 7(1-p)$$

$$-5p + 9 = 12p - 7.$$

$$17p = 16.$$

$$p = \frac{16}{17}.$$

$$A: \left(0, \frac{16}{17}, \frac{1}{17}, 0 \right)$$

$$B: \left(\frac{12}{17}, \frac{5}{17} \right)$$

(2) Inventory ⇒ Annual Demand (D) = 10^4 units.
 Ordering Cost (C_3) = Rs 10/order.
 Unit Price (p) = ₹ 20/unit.

Holding Cost = 20% of p .
 (C_1) $\Rightarrow (0.2 \times 20) = \text{Rs } 4 \text{ unit}$

Shortage / Backordering Cost (C_2) = 25% of unit price
 $\Rightarrow 0.25 \times 20$
 $\Rightarrow \text{₹ } 5 \text{ unit/yr}$

(i) Optimum no. of units to buy in one lot (q_0)

$$q_0 = \sqrt{\frac{2C_3 \cdot D}{C_1} \cdot \frac{C_1 + C_2}{C_2}}$$

$$q_0 = \sqrt{\frac{2 \times 10 \times 10^4}{4} \times \frac{9}{5}}$$

$$q_0 = \sqrt{90000} = 300 \text{ units}$$

(ii) Quantity allowed to be backordered

$$s = \left(\frac{C_1}{C_1 + C_2} \right) \times q_0$$

$$= \frac{4}{9} \times 300 = 133.33 \text{ units}$$

(iii) Max^m Quantity of inventory at any time (I_{max})

$$I_{\text{max}} = q_0 - s$$

$$= (300 - 133.33)$$

$$= 166.67 \text{ units}$$

Recommendation & Annual Cost saving

Total Cost without backordering (Standard EOQ):-

$$\begin{aligned}TC_{\text{no-backorder}} &= \sqrt{2 C_1 C_3 D} \\&= \sqrt{2 \times 4 \times 10 \times 10^4} \\&= \sqrt{8 \times 10^5} = \text{Rs } 894.43\end{aligned}$$

Total Cost with backordering:-

$$\begin{aligned}TC_{\text{backorder}} &= \sqrt{2 C_1 C_3 D \left(\frac{C_2}{C_1 + C_2} \right)} \\&= \sqrt{8 \times 10^5 \times \frac{5}{9}} \\&= \text{Rs } 666.67\end{aligned}$$

$$\begin{aligned}\text{Annual Cost Saving} &= \text{Rs } (894.43 - 666.67) \\&= \text{Rs } 227.76\end{aligned}$$

CO6:-

(a) Project Network, Critical Path & Float.
forward Pass:-
Earliest start (ES), Earliest finish (EF), Latest start (LS), Latest finish (LF)

$$A: ES = 0 \Rightarrow EF = 0 + 2 = 2$$

$$B: ES = 0 \Rightarrow EF = 0 + 1 = 1$$

$$C: ES = EF_A = 2 \Rightarrow EF = 2 + 3 = 5$$

(follows A)

$$D: ES = \max(EF_A, EF_B) = 2 \Rightarrow EF = 2 + 2 = 4$$

(follows A, B)

$$E: ES = \max(EF_C, EF_D) = 5 \Rightarrow EF = 5 + 1 = 6$$

(follows C, D)

F: $ES = \max(ES_B, ES_D) = 4 \Rightarrow EF = 4 + 3 = 7$
(follow B, D)

G: $ES = \max(ES_E, ES_F) = 7 \Rightarrow EF = 7 + 1 = 8$
(follow E, F)

Backward Pass (LS, LF):

G: $LF = 8 \Rightarrow LS = 8 - 1 = 7$

F: $LF = LS_G = 7 \Rightarrow LS = 7 - 3 = 4$
(Preceded)

E: $LF = LS_F = 7 \Rightarrow LS = 7 - 1 = 6$
(Preceded)

D: $LF = \min(LS_E, LS_F) = 4 \Rightarrow LS = 4 - 2 = 2$
(Preceded E, F)

C: $LF = LS_E = 6 \Rightarrow LS = 6 - 3 = 3$
(Preceded E)

B: $LF = \min(LS_D, LS_F) = 2$
(Preceded D, F)
 $LS = 2 - 1 = 1$

A: $LF = \min(LS_C, LS_D) = 2$
(Preceded C, D)

$LS = 2 - 2 = 0$

Ans

Total float (TF) = $(LS - ES)$ OR $(LF - EF)$
Free float (FF) = $\min(ES_{successor}) - EF$

Activity	Duration	ES	EF
A	2	0	2
B	1	0	1
C	3	2	5
D	2	2	4
E	1	5	6
F	3	4	7
G	1	7	8

Total float $\Rightarrow$ A: $0 - 0 = 0$
 B: $1 - 0 = 1$
 C: $3 - 2 = 1$ D: $2 - 2 = 0$
 E: $6 - 5 = 1$ F: $4 - 4 = 0$
 G: $7 - 7 = 0$

⑧

Activity	t_o	t_m	t_p
1-2	1	1	7
2-3	1	4	7
1-4	2	2	8
2-5	1	1	1
3-5	2	5	14
4-6	2	5	8
5-6	3	6	15

$$t_e = \frac{t_o + 4t_m + t_p}{6}$$

$$\sigma^2 = \left(\frac{t_p - t_o}{6} \right)^2$$

Path & Project Lengths:

Path	
1-2-5-6	2+1+7 = 10 wks.
1-3-5-6	4+6+7 = 17 wks. → (Critical)
1-4-6	3+5 = 8 wks.

$$P_e = 17 \text{ wks.}$$

$$\text{Project Variance } (\sigma_{\text{proj}}^2) = 1 + 4 + 4 = 9$$

$$(\text{1-3, 3-5, 5-6}) \quad \sigma = \sqrt{9} = 3$$

① At least 4 wks earlier than expected.

$$\text{Target Time } (T_1) \leq 17 - 4 = 13 \text{ wks.}$$

$$Z = \frac{T_1 - P_e}{\sigma} = \frac{13 - 17}{3} = -1.33$$

$$Z \leq -1.33 \rightarrow P = 0.0918$$

$$\downarrow$$

$$9.18\%$$

② No more than 4 wks later than expected.

$$T_1 \leq 17 + 4 = 21 \text{ wks.}$$

$$Z = \frac{21 - 17}{3} = \frac{4}{3} = 1.33$$

$$Z \leq 1.33$$

$$\downarrow$$

$$1 - 0.0912 = 0.9082$$

$$\downarrow$$

$$90.82\%$$

① $(M/M/1) \rightarrow$ Queuing Model.

$M/M/1 : \infty / F / CFS$.

Poisson arrivals at rate λ ,

Exponential service at rate μ .

Prob ≈ 1 $\xrightarrow{\text{arrival}}$ $\Delta t \approx \lambda \Delta t$.

Prob ≈ 0 $\xrightarrow{\text{''}}$ $\Delta t \approx 1 - \lambda \Delta t$.

Prob ≈ 1 $\xrightarrow[\text{Completion}]{\text{service}}$ $\Delta t \approx \mu \Delta t$.

~~Prob ≈ 0 $\xrightarrow{\text{''}}$ $\Delta t \approx 1 - \mu \Delta t$.~~

Case-01: System can be in state 'n' at time $t + \Delta t \rightarrow$
 $(n, 1)$.

① State n , 0 arr., 0 depl.

$$P_n(t) \cdot (1 - \lambda \Delta t) (1 - \mu \Delta t) \\ \approx P_n(t) (1 - (\lambda + \mu) \Delta t)$$

② State $n-1$, 1 arr., 0 depl.

$$P_{n-1}(t) (\lambda \Delta t) (1 - \mu \Delta t)$$

③ State $n+1$, 0 arr., 1 depl.

$$P_{n+1}(t) (1 - \lambda \Delta t) (\mu \Delta t) \\ \approx P_{n+1}(t) \cdot \mu \Delta t$$

① + ② + ③

$$P_n(t + \Delta t) = [P_n(t) (1 - (\lambda + \mu) \Delta t)] + P_{n-1}(t) \lambda \Delta t \\ + P_{n+1}(t) \mu \Delta t$$

$$\frac{P_n(t+\Delta t) - P_n(t)}{\Delta t} = -(\lambda + \mu)P_n(t) + \lambda P_{n-1}(t) + \mu P_{n+1}(t)$$

Limit $\Delta t \rightarrow 0$,

$$P_n'(t) = \lambda P_{n-1}(t) + \mu P_{n+1}(t) - (\lambda + \mu)P_n(t)$$

Case - 2:-

($n=0$)

In state 0 at time $t + \Delta t$:-

(i) 0 arr. & state > 0 (No dep.)
 $P_0(t) \cdot (1 - \lambda(\Delta t))$

(ii) State 1, 0 arr, 1 dep. (1 arr)
 $P_1(t) \cdot \mu(\Delta t)$

$$P_0(t+\Delta t) = P_0(t)(1 - \lambda(\Delta t)) + P_1(t)\mu\Delta t$$

$$\frac{P_0(t+\Delta t) - P_0(t)}{\Delta t} = -\lambda P_0(t) + \mu P_1(t)$$

$\Delta t \rightarrow 0$

$$P_0'(t) = \mu P_1(t) - \lambda P_0(t)$$